Short-pulse sequences as local probes for superconducting-qubit frequency calibration

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I. INTRODUCTION

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II. SYSTEM AND CALIBRATION TASK

A. Flux-tunable transmon and calibration objective

We consider a flux-tunable transmon qubit [**?**] controlled through a calibrated flux-bias coordinate Φ. In the transmon regime, the flux dispersion of the qubit angular transition frequency is approximately given by [**?**]

$$\hbar\omega_{01}(\Phi) \simeq \sqrt{8E_C E_J(\Phi)} - E_C, \tag{1}$$

Equation (1) describes the flux dependence of the transition frequency and thus provides the physical basis for frequency calibration through flux-bias tuning. The calibration objective is to find a flux bias $\Phi_\star$ such that

$$|\omega_{01}(\Phi_\star) - \omega_{\mathrm{tar}}| \leq \epsilon_\omega, \tag{2}$$

where ω_{tar} is the target angular transition frequency and ϵ_ω is the prescribed tolerance.

In practice, a previously characterized frequency–flux map may no longer be accurate at the target tolerance because of slow shifts in the effective flux offset or device parameters[**? ?**]. We therefore formulate the calibration without assuming an up-to-date quantitative map $\omega_{01}(\Phi)$: the controller operates in the calibrated flux coordinate and obtains the local frequency information required for bias updates directly from measurements.

Figure 1(a) summarizes this calibration setting. The full dispersion illustrates the physical tunability of the qubit, while the inset shows a local trajectory from the seed bias toward the target frequency.

B. Ramsey and bipartite short-pulse probes

At each calibration iteration, the qubit is interrogated at a fixed flux bias by a control sequence followed by population readout. We refer to this pulse-and-readout operation as a measurement probe. We consider two types of probes: Ramsey probes and bipartite short-pulse probes. The driven dynamics of both probes are described by an effective two-level Hamiltonian. In a frame rotating at the drive frequency ω_d and under the rotating-wave

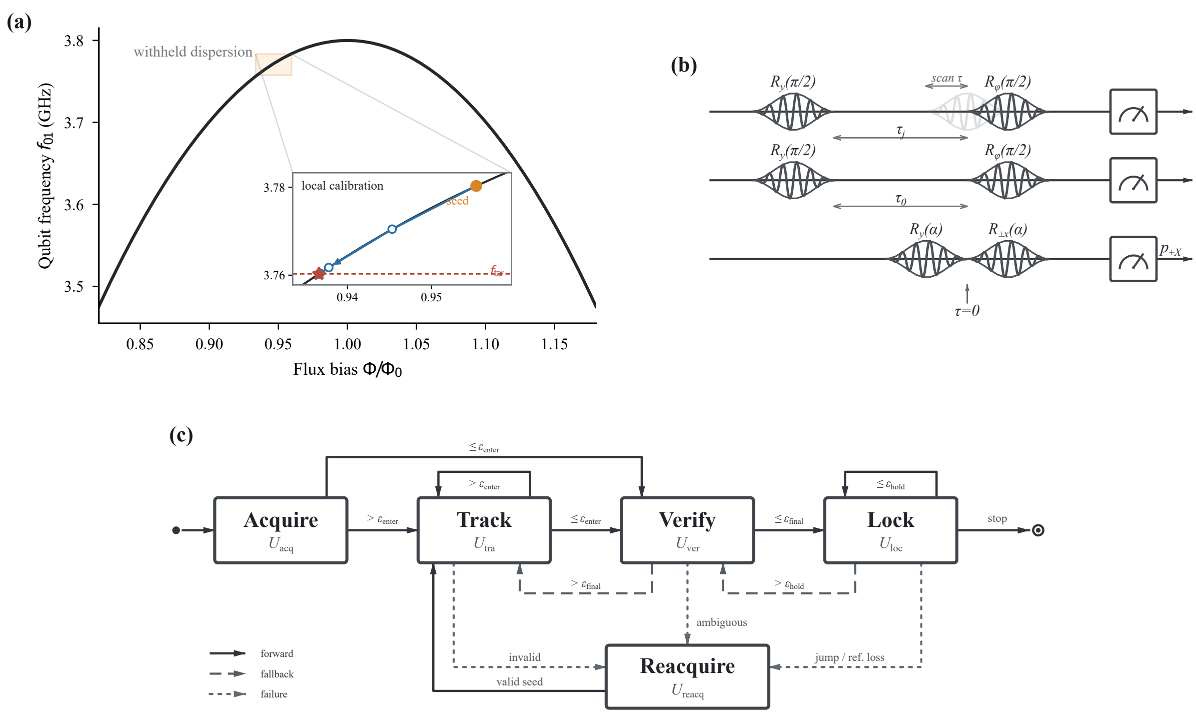

FIG. 1. Schematic overview of the physical system, measurement probes, and calibration protocol. (a) Qubit frequency dispersion, with an inset showing the seed bias, target frequency, and successive points of the local calibration trajectory. The complete dispersion is withheld from the calibration controller. (b) Timing diagrams for scanned Ramsey, fixed-delay Ramsey, and the bipartite short-pulse probe. (c) The Acquire–Track–Verify–Lock protocol, with Reacquire recovery and a terminal safe hold. Only Track changes the flux bias; Verify freezes the candidate bias for independent confirmation, and Lock performs long-term monitoring. Solid, dashed, and dotted arrows denote forward, fallback, and failure transitions, respectively.

approximation [? ?],

$$\frac{H(t,\Phi)}{\hbar} = \frac{\Delta(\Phi)}{2}\sigma_z + \frac{\Omega(t)}{2}\left[\cos\varphi(t)\sigma_x + \sin\varphi(t)\sigma_y\right],$$
$$\Delta(\Phi) = \omega_{01}(\Phi) - \omega_d. \tag{3}$$

Here $\Delta(\Phi)$ is the angular-frequency detuning, while $\Omega(t)$ and $\varphi(t)$ are the drive envelope and phase. The detuning term connects the unknown transition frequency to the driven dynamics. Ramsey and short-pulse probes use the distinct control sequences shown in Fig. 1(b) to infer this detuning from population readout. once $\widehat{\Delta}$ is obtained, the transition frequency follows as $\widehat{\omega}_{01} = \omega_d + \widehat{\Delta}$.

We first consider the Ramsey probe, which uses two nominal $\pi/2$ pulses separated by a free-evolution interval τ [?]. For an ideal Ramsey sequence with analysis phase φ and Ramsey-fringe visibility C, the measured population can be written as

$$p_R(\Delta;\tau,\varphi) = \frac{1}{2}\left[1 + C\cos(\Delta\tau + \varphi)\right], \tag{4}$$

We use scanned Ramsey, which samples multiple values of τ, for wide-range frequency estimation. As a local baseline, we instead fix $\tau = \tau_0$ and use two orthogonal analysis phases to recover the detuning from measured quadratures within the unambiguous phase range [?].

We next turn to the bipartite short-pulse probe. Unlike the Ramsey probe, it contains no separate free-evolution interval and instead applies a common preparation rotation of angle α immediately followed by one of two analysis rotations whose phases differ by π [?]:

$$R_y(\alpha) \longrightarrow R_{+x}(\alpha) \quad \text{or} \quad R_y(\alpha) \longrightarrow R_{-x}(\alpha). \tag{5}$$

Here $\alpha = \int \Omega(t)\,\mathrm{d}t$ is determined from the pulse envelope and duration. Varying α changes the differential detuning response and therefore the response coefficients and validated operating range introduced in Sec. III.

We refer to these two sequences as the $+X$ and $-X$ branches and denote their final excited-state populations by p_{+X} and p_{-X}, respectively. We define the corresponding differential observable as

$$p_{\mathrm{d}} = \frac{p_{+X} - p_{-X}}{2}. \tag{6}$$

Detuning acts during the finite control pulses and changes the measured branch populations. Their difference provides a first-order-sensitive local frequency observable when

$$\left.\frac{\partial p_{\mathrm{d}}}{\partial \Delta}\right|_{\Delta=0} \neq 0. \tag{7}$$

This condition is necessary for local inversion but does not guarantee an unambiguous response over an unrestricted detuning range. Section III quantifies the local response and establishes its validated operating interval.

C. Calibration workflow

The local-identifiability condition applies at a fixed bias, whereas a bias update can move the qubit outside the validated short-pulse range. We therefore embed the local probe in the ACQUIRE–TRACK–VERIFY–LOCK workflow shown in Fig. 1(d), supplemented by REACQUIRE and terminal SAFESTOP states. This architecture is motivated by automated qubit calibration and closed-loop frequency feedback [**? ? ?**].

At iteration n, the measurement detuning and target residual are

$$\Delta_n = \omega_{01}(\Phi_n) - \omega_{d,n}, \qquad r_n = \omega_{01}(\Phi_n) - \omega_{\mathrm{tar}}. \tag{8}$$

The detuning Δ_n determines whether the local probe remains valid, whereas r_n determines whether the target has been reached. A flux-bias update changes the physical working point to reduce r_n; a drive-reference update controls the detuning of the next measurement without changing the target.

For both ACQUIRE and TRACK, an estimate becomes a target candidate when

$$|\widehat{r}_{x,k}| + z_{1-\beta}\sigma_{x,k} \le \epsilon_{\mathrm{enter}}, \qquad x \in \{\mathrm{acq}, \mathrm{tra}\}, \tag{9}$$

where $\sigma_{x,k}$ is the uncertainty of the estimated target residual and $z_{1-\beta}$ is the prescribed confidence multiplier.

In ACQUIRE, scanned Ramsey estimates the current qubit frequency and initializes the drive. An estimate satisfying Eq. (9) is routed directly to VERIFY; otherwise, it seeds TRACK. Only TRACK updates the flux bias, using the short-pulse frequency estimate while updating the drive reference to keep subsequent measurements within the validated range. When a TRACK estimate satisfies Eq. (9), the bias is frozen and passed to VERIFY. A failed local-validity check instead enters REACQUIRE.

VERIFY uses independent Ramsey measurements at the frozen bias. Calibration is confirmed only after N_{verify} consecutive measurements satisfy

$$|\widehat{r}_{\mathrm{ver},j}| + z_{1-\beta}\sigma_{\mathrm{ver},j} \le \epsilon_{\mathrm{final}}. \tag{10}$$

A correctable verification error returns to TRACK, whereas a lost frequency reference or an error outside the local recovery range enters REACQUIRE.

After successful verification, LOCK holds the verified bias and uses periodic low-cost short-pulse measurements to monitor drift. Detected drift or a scheduled audit returns the protocol to VERIFY; a large frequency jump or loss of the local reference enters REACQUIRE. REACQUIRE repeats the wide-range acquisition and applies the same routing rule as ACQUIRE. Cancellation, budget exhaustion, or a safety interlock enters SAFESTOP. Detailed thresholds and schedules are specified in Appendix A.

III. SHORT-PULSE FREQUENCY INFERENCE AND CALIBRATION

We now specify the local TRACK branch introduced in Sec. II.C. A usable tracking iteration must convert the differential population into a frequency estimate, move the physical working point toward ω_{tar}, and keep the next measurement within the validated response interval as the qubit frequency changes. We therefore derive the short-pulse differential response, construct its nonlinear local inverse together with an explicit validity boundary, use the resulting frequency estimate in a bounded flux-bias update, and update the drive reference to preserve the local measurement condition. Together, these elements specify how, and within what limits, short-pulse frequency information can be incorporated into closed-loop flux-bias calibration.

A. Short-pulse differential response

To characterize how detuning is encoded in Eq. (6), we regard the differential population as a functional of a time-dependent detuning $\Delta(t)$. Following the general framework of linear and nonlinear response theory [**? ?**], we expand the differential population around the resonant control evolution as

$$p_{\mathrm{d}}[\Delta] = p_{\mathrm{d}}^{(0)} + \sum_{m=1}^{\infty} \frac{1}{m!} \int_0^T \mathrm{d}t_1 \cdots \int_0^T \mathrm{d}t_m \times k_m(t_1, \ldots, t_m) \prod_{j=1}^{m} \Delta(t_j). \tag{11}$$

Here $k_m(t_1, \ldots, t_m)$ is the symmetrized mth-order detuning-response kernel of the complete bipartite short-pulse probe. It quantifies the joint contribution of detuning at the times $t_1, \ldots, t_m$ to the final differential population, and T is the total sequence duration. The response-function derivation, including time ordering and nested commutators, is given in the Supplemental Material.

For scalar local frequency estimation, we restrict attention to the quasi-static regime in which detuning variations during a single short sequence are negligible, so that $\Delta(t) \simeq \Delta$. This approximation does not preclude drift between repetitions or feedback iterations.

For the ideal $+X/-X$ probe defined in Eq. (5), detuning reversal interchanges the branch responses, $p_{+X}(-\Delta) = p_{-X}(\Delta)$. The differential combination therefore satisfies $p_{\mathrm{d}}(-\Delta) = -p_{\mathrm{d}}(\Delta)$. In particular, $p_{\mathrm{d}}^{(0)} = 0$, and all even-order terms in Eq. (11) vanish. The local response can thus be written as

$$p_{\mathrm{d}}(\Delta) = G_1\Delta + \frac{G_3}{3!}\Delta^3 + \mathcal{O}(\Delta^5), \tag{12}$$

where G_1 and G_3 are the first and third derivatives of p_d with respect to Δ at resonance. The coefficient G_1 determines the local sensitivity, while G_3 gives the leading nonlinear correction.

For the experimental implementation, we separate the response measurements into calibration and validation data. From the response-calibration data, we obtain G_1 and G_3 through an offline, measurement-based kernel calibration. Calibrated Virtual-Z phase insertions are applied at selected times within the two control branches, and finite differences of the resulting branch populations sample the detuning-response derivatives. In the quasi-static limit, the integrated coefficients are

$$\begin{aligned} G_1 &= \int_0^T k_1(t)\ \mathrm{d}t, \\ G_3 &= \iiint_{[0,T]^3} k_3(t_1,t_2,t_3)\ \mathrm{d}t_1\,\mathrm{d}t_2\,\mathrm{d}t_3. \end{aligned} \tag{13}$$

The Virtual-Z operations follow Ref. [**?**]. The finite-difference construction and three-time sampling scheme used to obtain the response kernels are detailed in the Supplemental Material.

Experimental imperfections may nevertheless produce a nonzero measured differential signal at resonance. We therefore define

$$\widetilde{p}_\mathrm{d} = p_\mathrm{d}^\mathrm{meas} - b_0, \tag{14}$$

where b_0 is the zero-detuning intercept estimated from the response-calibration data and held fixed thereafter. This operation corrects a differential-population offset; it does not shift the frequency reference.

For comparison, an odd-polynomial fit to the corrected quasi-static response provides a separate check of the integrated coefficients but is not used to generate the coefficients supplied to the main feedback protocol. Once G_1, G_3, and b_0 have been fixed, separately acquired response-validation data are used to establish Δ_val and to assess residual symmetry breaking through $p_\mathrm{even}(\Delta) = [\widetilde{p}_\mathrm{d}(\Delta) + \widetilde{p}_\mathrm{d}(-\Delta)]/2$. The even component is treated as a diagnostic of model mismatch rather than absorbed into the ideal odd response model. This response validation is distinct from the independent frequency verification used to confirm calibration success in Sec. II.C.

B. Local frequency estimator

The first-order frequency estimate follows from the local slope:

$$\widehat{\Delta}^{(1)} = \frac{\widetilde{p}_\mathrm{d}}{G_1}, \qquad \widehat{\omega}_{01}^{(1)} = \omega_d + \widehat{\Delta}^{(1)}. \tag{15}$$

Its truncation bias increases as the probe moves away from resonance. This estimate provides the initial value for the cubic inversion and the first-order reference used to quantify nonlinear-estimation bias in Sec. IV.

The main local estimator retains the third-order response and solves

$$G_1\widehat{\Delta}^{(3)} + \frac{G_3}{6}\left(\widehat{\Delta}^{(3)}\right)^3 = \widetilde{p}_\mathrm{d}, \tag{16}$$

with $\widehat{\omega}_{01}^{(3)} = \omega_d + \widehat{\Delta}^{(3)}$.

When $G_1G_3 < 0$, the cubic response has stationary points at

$$|\Delta_\mathrm{fold}| = \sqrt{-\frac{2G_1}{G_3}}. \tag{17}$$

Beyond these points, the cubic response is no longer one-to-one, and the same differential population can correspond to multiple detunings. The response fold is therefore a mathematical boundary of the truncated cubic model, not an experimentally established operating range.

We define the validated local operating interval by

$$|\Delta| \le \Delta_\mathrm{val}, \qquad \Delta_\mathrm{val} < |\Delta_\mathrm{fold}| \quad (G_1G_3 < 0). \tag{18}$$

The value of Δ_val is established from the held-out response-validation data using predefined estimator-error and confidence criteria. It remains necessary when the cubic model has no real fold because model truncation, measurement noise, and experimental imperfections can restrict the usable response before a mathematical turnover occurs. Failure of numerical convergence, branch continuity, or the validated-range check rejects the estimate and triggers Reacquire rather than extrapolation of the local inverse.

C. Track initialization and flux-bias update

Within the workflow defined in Sec. II.C, an Acquire or Reacquire measurement supplies the seed $(\Phi_\mathrm{seed}, \widehat{\omega}_{01,\mathrm{seed}}, \sigma_{\omega,\mathrm{seed}})$. For a seed routed to Track, the acquired bias–frequency pair serves as the first point of the secant initialization. It sets the initial drive reference $\omega_{d,\mathrm{seed}} = \widehat{\omega}_{01,\mathrm{seed}}$ and the residual $\widehat{r}_\mathrm{seed} = \widehat{\omega}_{01,\mathrm{seed}} - \omega_\mathrm{tar}$.

To obtain the second bias–frequency point required for the secant estimate, the controller proposes the bounded one-sided step

$$\Phi_1^\mathrm{prop} = \Phi_\mathrm{seed} - \chi_\Phi\,\mathrm{sgn}(\widehat{r}_\mathrm{seed})\delta\Phi_\mathrm{blind}, \tag{19}$$

where $\chi_\Phi = \mathrm{sgn}(\partial\omega_{01}/\partial\Phi) \in \{-1,+1\}$ is the known local monotonic direction. The applied value Φ_1 is restricted to $[\Phi_\mathrm{min}, \Phi_\mathrm{max}]$. Let $\Delta_\mathrm{blind}^\mathrm{max}$ be a validated upper bound on the frequency displacement produced by the applied blind step. The initialization is locally admissible only if

$$z_{1-\beta}\sigma_{\Delta,\mathrm{seed}} + \Delta_\mathrm{blind}^\mathrm{max} \le \Delta_\mathrm{val}, \tag{20}$$

where $\sigma_{\Delta,\mathrm{seed}}$ includes the uncertainty of the acquired frequency and the implemented seed reference. This condition reserves the detuning margin required for the first short-pulse measurement at Φ_1. Before a secant sensitivity is available, the first short-pulse measurement at Φ_1 uses $\omega_{d,1} = \widehat{\omega}_{01,\mathrm{seed}}$ and returns $\widehat{\omega}_{01,1}$. Together with the scanned-Ramsey seed, this measurement provides the initial secant sensitivity; its uncertainty is propagated from both frequency estimates.

Subsequent sensitivities are calculated from the two most recent TRACK estimates,

$$\widehat{S}_{\Phi,n} = \frac{\widehat{\omega}_{01,n} - \widehat{\omega}_{01,n-1}}{\Phi_n - \Phi_{n-1}}, \qquad n \geq 2. \tag{21}$$

Thus, the controller estimates the local flux-to-frequency sensitivity without evaluating the quantitative transmon dispersion or its derivative. Only sensitivities with the prescribed sign, magnitude, and uncertainty are accepted; failure of these checks reports the local-invalid outcome defined in Sec. II.C.

At an accepted iteration, the controller forms $\widehat{r}_n = \widehat{\omega}_{01,n} - \omega_{\mathrm{tar}}$ and applies the target-entry test in Eq. (9). If the candidate is not routed to VERIFY, the next bias proposal is

$$\Phi^{\mathrm{prop}}_{n+1} = \Phi_n - \eta \frac{\widehat{r}_n}{\widehat{S}_{\Phi,n}}, \tag{22}$$

where $\eta \in (0, 1]$ is the damping factor. The applied bias change is limited to $\delta\Phi_{\max}$ in magnitude, and the resulting bias is restricted to $[\Phi_{\min}, \Phi_{\max}]$. The blind-step bound, uncertainty propagation, sensitivity thresholds, and limiting rules are specified in Appendix A.

Equation (22) changes the physical working point to reduce the residual relative to the fixed target. Maintaining the next short-pulse measurement within its validated detuning interval requires the separate drive-reference update developed below.

D. Drive-reference tracking and range maintenance

If the drive reference is held fixed after a flux-bias update, the next measurement detuning is $\Delta^{\mathrm{fixed}}_{n+1} = \omega_{01}(\Phi_{n+1}) - \omega_{d,n}$. The short-pulse probe must then span the accumulated frequency displacement as the physical bias moves and can leave its validated interval even when the qubit approaches the target.

During TRACK, we instead use the measured secant sensitivity to set the next drive reference,

$$\omega_{d,n+1} = \widehat{\omega}_{01,n} + \widehat{S}_{\Phi,n} \left(\Phi_{n+1} - \Phi_n\right). \tag{23}$$

The right-hand side predicts the transition frequency at Φ_{n+1} using only measurements available through iteration n; it does not evaluate Eq. (1). Consequently, the next measurement detuning is the one-step prediction error rather than the accumulated frequency displacement.

Let $\sigma_{\mathrm{pred},n+1}$ denote the propagated uncertainty of this frequency prediction, including the uncertainties of $\widehat{\omega}_{01,n}$, $\widehat{S}_{\Phi,n}$, and the applied bias step. Because Eq. (23) centers the predicted detuning at zero, the next short-pulse measurement is admitted only if

$$z_{1-\beta}\sigma_{\mathrm{pred},n+1} + \Delta_{\mathrm{guard}} \leq \Delta_{\mathrm{val}}, \tag{24}$$

where $\Delta_{\mathrm{guard}} \geq 0$ reserves a fixed margin for residual prediction and actuation errors not included in the propagated uncertainty. The initial blind step is governed instead by Eq. (20).

After an admitted measurement, the inferred detuning must satisfy

$$|\widehat{\Delta}_{n+1}| + z_{1-\beta}\sigma_{\Delta,n+1} \leq \Delta_{\mathrm{val}}. \tag{25}$$

The premeasurement condition protects against an unreliable drive prediction, whereas the postmeasurement condition checks consistency with the validated short-pulse interval. Failure of either condition, inverse convergence or branch continuity, or the sensitivity check reports the local-invalid outcome defined in Sec. II.C.

When these checks are satisfied, the frequency estimate is passed to the next flux-bias update; satisfaction of Eq. (9) instead reports a target-entry outcome. The uncertainty propagation and the choice of Δ_{guard} are specified in Appendix A.

Thus, Eq. (22) moves the physical working point, whereas Eq. (23) only recenters the next measurement. The calibration residual remains referenced to the fixed ω_{tar}.

IV. RESULTS

V. DISCUSSION

VI. CONCLUSION

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Appendix A: Numerical parameters and cost accounting